

Causal Entropy: A Graph-Theoretic Framework for the Origin of Low Initial Entropy

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Abstract

We propose the **Causal Entropy Hypothesis** (CEH): a theoretical framework in which entropy is not fundamentally a measure of disorder or missing information, but rather the *realized causal unfolding of energy* within a directed causal graph $G = (N, E)$. Under this reinterpretation, the classically puzzling low entropy of the early universe follows naturally from first principles: the initial causal graph was nearly edge-free ($|E| \approx 0$), implying minimal realized interactions and therefore minimal causal entropy $S_C \rightarrow S_{\min}$. We formalize five foundational axioms governing energy conservation, causal priority, graph ontology, latent structure, and irreducible uncertainty. From these axioms we derive a composite causal entropy functional incorporating graph size, epistemic uncertainty, latent causal pressure, and energy flux. We establish an entropy growth law, redefine thermodynamic equilibrium as a *rate condition* rather than a state condition, and introduce an Emergent Compression Principle whereby stable graphs contract into super-nodes that seed new causal cycles. The framework is applied to black hole thermodynamics, re-interpreting Bekenstein–Hawking entropy as extreme causal path convergence. Four testable predictions are stated. The CEH does not replace thermodynamics; it proposes a deeper causal substrate beneath it.

Keywords: causal entropy; causal graphs; initial entropy; hidden variables; emergent complexity; black hole entropy; graph ontology

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1. Introduction

The second law of thermodynamics is among the most empirically robust results in physics: the total entropy of an isolated system does not decrease over time. Yet this law is deeply asymmetric in time, and its application to cosmology raises a question that standard statistical mechanics does not answer from first principles:

Why was the entropy of the early universe so extraordinarily low?

The Penrose–Boltzmann account [1] holds that the initial state of the universe was one of very special low entropy, but treats this as a boundary condition rather than an explained consequence. Carroll and Chen [2] argue for a cosmological scenario in which low-entropy pockets arise spontaneously, but the origin of the *first* low-entropy state remains an open problem.

This paper proposes a different approach. Rather than asking “*why was the entropy low?*” as a question about statistics or probability over phase space, we ask: “*what is entropy, more fundamentally?*” We argue that entropy is the *footprint of realized causal structure* — the cumulative record of interactions that have actually occurred, edges that have actually formed in the causal graph of reality. Under this interpretation, low initial entropy is not a puzzle to be explained by special initial conditions; it is the *tautological consequence* of there being, at the beginning, almost no realized causal interactions at all.

1.1 Scope and Claims

The Causal Entropy Hypothesis (CEH) makes the following positive claims:

- (i) Entropy is fundamentally the realized causal unfolding of energy, not disorder.
- (ii) The universe’s initial low entropy reflects a nearly empty causal graph, not a special fine-tuned microstate.
- (iii) A composite causal entropy functional S_C can be constructed that recovers qualitative thermodynamic behavior while incorporating latent (hidden) causal structure.
- (iv) Thermodynamic equilibrium is a *rate condition*: entropy growth approaches zero, but uncertainty never vanishes.
- (v) Stable causal configurations compress into higher-level nodes, enabling recursive hierarchical organization of reality.

We do not claim to derive the CEH from an existing axiomatic system. We offer it as a novel conceptual and mathematical framework, state its axioms explicitly, derive its principal consequences, and identify falsifiable predictions. Reconciliation with quantum field theory and general relativity is left as future work.

2. Foundational Axioms

We begin with six axioms that constitute the ontological commitments of the CEH.

Axiom 1 (Energy Conservation). Total energy is conserved:

$$E_{\text{total}} = \text{const.}$$

Energy is neither created nor destroyed; it is only redistributed through interactions.

Axiom 2 (Interaction Before Observation). Cause precedes interaction; interaction precedes effect; effect precedes observation; observation precedes data:

$$\text{Cause} \longrightarrow \text{Interaction} \longrightarrow \text{Effect} \longrightarrow \text{Observation} \longrightarrow \text{Data}.$$

Data does not produce causes. Data is the *footprint* of a causal chain that preceded it. In the absence of an interaction, there is no energy redistribution, no effect, no observation, and no data. In a physical system such as an engine: combustion (burn) occurs first; CA50 is the effect; sensor data is the downstream footprint. The causal discovery algorithm (DIFC) arrives last — it does not create the causal structure it finds.

Axiom 3 (Graph Ontology). Physical reality can be faithfully represented as a directed causal graph $G = (N, E)$, where N is the set of nodes (physical entities or configurations) and E is the set of directed causal edges (realized interactions). An *observable state* is a snapshot projection of G ; the graph itself is the fundamental object.

Axiom 4 (Entropy as Realized Causal Structure). Entropy is not a measure of disorder, randomness, or missing information about a static configuration. Entropy is the *footprint* of realized causal activity:

$$S \sim \text{realized causal unfolding of energy.}$$

More precisely, entropy reflects: (i) realized causal edges formed, (ii) energy redistribution that has occurred, and (iii) latent causal organization not yet observable. Where no interactions have been realized, no entropy footprint exists. Entropy grows

only as causal structure unfolds.

Remark 2.1. Axiom 4 is the central ontological commitment of CEH. It distinguishes the framework from both Boltzmann entropy (which counts microstates) and Shannon entropy (which measures information deficit). Boltzmann and Shannon entropies are not rejected; they are re-interpreted as special projections of the causal entropy footprint onto phase-space and information-theoretic substrates respectively.

Axiom 5 (Latent Reality). No observed graph G is complete. There exist hidden causal nodes and edges $H \notin G$ such that $H \in \mathcal{R}$ (reality). The latent complement $\mathcal{R} \setminus G$ exerts measurable pressure on observable dynamics.

Axiom 6 (Irreducible Uncertainty). Perfect epistemic certainty is unattainable for any finite observer:

$$U > 0 \quad \text{always (for any observer embedded in a graph it cannot fully observe).}$$

Uncertainty U may approach a minimum stable value but never reaches zero.

Critical scope condition: U is an observer-dependent, operationally defined quantity. It requires an observer, observations, and data to be meaningful. In a state where no interactions have occurred ($I = 0$), there are no observations and no data — therefore U is not operationally defined. Axiom 6 applies to any observer *after* at least one interaction has been realized. It does not apply at $I = 0$, and consequently $U > 0$ cannot serve as the mechanism that *initiates* the first interaction.

Remark 2.2. This scope condition is essential. A prior version of this framework incorrectly used $U > 0$ as the engine that generates the first causal fluctuation. That argument is invalid: U presupposes an observer with data, and data presupposes interactions (Axiom 2). The correct answer to the first-interaction problem is provided by the Recursive Reality Principle (Section 8.3).

3. The Initial Universe Under CEH

3.1 Initial Causal Graph State

Under Axiom 3, the state of the universe at any epoch is characterized by the properties of its causal graph $G_t = (N_t, E_t)$. We define realized energy redistribution as the cumulative energy flux integral:

$$I(t) = \int_0^t \left| \frac{dE_{\text{flow}}}{dt} \right| dt'. \quad (1)$$

The foundational claim about the initial universe is the following:

Hypothesis 3.1 (Constrained Initial State). At the initial epoch $t \rightarrow 0$, the universe was in a highly constrained state in which energy redistribution had not yet been realized:

$$I(0) \approx 0.$$

Under the Recursive Reality Principle (Section 8.3), this hypothesis gains a structural interpretation that removes it from the status of a bare postulate. If reality organizes recursively — each stable causal graph compressing into a super-node that seeds the next graph — then the “initial” state of our observable causal level is simply the state of a newly formed super-node at the beginning of a new cycle:

$$G_{n-1} \longrightarrow \text{equilibrium} \longrightarrow N_{\text{super}} = G_n(t = 0).$$

At the moment of formation, N_{super} has no realized edges in G_n . Therefore $I_n(0) \approx 0$ is not an arbitrary fine-tuned choice — it is the *expected state of any super-node at the start of a new causal cycle*. The question “why was the initial state constrained?” is answered: because it was the compressed output of the previous level’s equilibration. *Remark 3.2.* This interpretation does not eliminate all regress — it replaces a single mysterious beginning with a recursive structure that has no absolute beginning. Whether this is philosophically preferable is a separate question. What it does achieve is: the initial constrained state is no longer an arbitrary boundary condition; it is the structural consequence of the Recursive Reality Principle applied one level up.

From Hypothesis 3.1, the near-empty causal graph follows as a *consequence*, not an independent assumption. By Axiom 2, every causal edge requires a realized interaction; if no energy redistribution has occurred ($I \approx 0$), then no interactions have taken place, and therefore no causal edges exist:

$$I \approx 0 \implies \text{no realized interactions} \implies |E| \approx 0. \quad (2)$$

The derivation chain is therefore:

$$\underbrace{\text{Constrained State}}_{\text{assumed}} \longrightarrow I \approx 0 \longrightarrow \text{no realized interactions} \longrightarrow |E| \approx 0 \longrightarrow S_{\mathcal{C}} \rightarrow S_{\min}.$$

This is a non-trivial structural claim. The empty causal graph is not posited

directly; it is derived from the more primitive condition of minimal realized energy redistribution. The question “why was the causal graph empty?” therefore has an internal answer within CEH: because no interactions had yet occurred. The question that remains open — and is explicitly acknowledged as such — is one level deeper: why was the initial state so highly constrained?

3.2 Reframing the Low-Entropy Puzzle

Hypothesis 3.3 (Low Initial Entropy). Thermodynamic entropy was low in the early universe because the realized causal structure of the universe was minimal. Entropy, under CEH, is not identical to causal structure; it is the *footprint* of realized causal structure, energy redistribution, and latent organization. The claim is therefore:

$$S_C \propto |E|, \quad |E| \rightarrow 0 \implies S_C \rightarrow S_{\min}.$$

Combined with the derivation chain (2), this gives:

$$I(0) \approx 0 \implies |E| \approx 0 \implies S_C \rightarrow S_{\min}.$$

Under CEH, the question “why was entropy low?” decomposes into two sub-questions of different status:

- (i) *Why was the causal graph nearly empty?* — **Answered within CEH:** because no energy redistribution had yet been realized ($I \approx 0$), and by Axiom 2 no interactions had therefore occurred. The empty graph is a *consequence* of minimal realized interaction, not an independent assumption.
- (ii) *Why was the initial state so highly constrained?* — **Open problem:** CEH does not yet derive the constrained initial state from a deeper principle. This is the remaining open assumption of the framework, and it is shared with standard cosmology, which also accepts initial low-entropy conditions without deriving them from first principles [1, 2].

The strongest honest statement of CEH’s explanatory contribution is therefore:

CEH derives the near-empty initial causal graph from minimal realized energy redistribution. CEH does not yet derive why the initial state was so highly constrained. This latter question remains open.

What the reframing does achieve is a shift in *explanatory substrate*: the puzzle moves from statistical mechanics (special microstate in phase space) to causal graph

theory (minimal realized interaction). This shift is non-trivial if and only if the causal-entropy functional S_C can be computed numerically and shown to reproduce known thermodynamic behavior — which remains the critical empirical test.

4. Causal Entropy: Formal Definition

4.1 Motivation

Classical Boltzmann entropy $S = k_B \ln \Omega$ counts microstates consistent with a macrostate. Shannon entropy $H = -\sum p_i \ln p_i$ measures information deficit. Both treat entropy as a property of an observer’s knowledge about a static configuration. CEH instead treats entropy as a *process variable*: it grows with the actual unfolding of causal events.

A critical distinction must be stated explicitly: CEH does *not* identify entropy *with* causal structure. Entropy is proposed to be the **footprint** of realized causal structure — the cumulative observable record left by interactions, energy redistribution, and latent causal organization. The causal graph G is the substrate; S_C is the imprint that substrate leaves on measurable quantities. This distinction matters: a highly concentrated system (e.g. a black hole) can have very high causal convergence and therefore very high entropy, which is consistent with CEH and with observation. The framework does not predict that “structured = low entropy”; it predicts that entropy grows with *realized* causal unfolding, whatever form that takes.

We require an entropy functional that:

- (a) grows as the causal graph acquires edges ($|E| \uparrow$),
- (b) increases when epistemic uncertainty increases ($U \uparrow$),
- (c) reflects unresolved latent structure ($L \uparrow$),
- (d) grows with realized energy flux.

4.2 The Causal Entropy Functional

Definition 4.1 (Causal Entropy). The **causal entropy** of a system represented by graph $G = (N, E)$ at time t is:

$$S_C(t) = \alpha \ln(1 + |E|) + \beta U + \gamma L + \delta \int_0^t \left| \frac{dE_{\text{flow}}}{dt} \right| dt' \quad (3)$$

where:

$$\begin{aligned}
|E| &= \text{number of realized causal edges in } G, \\
U &= \text{epistemic uncertainty (Axiom 6)}, \\
L &= \text{latent causal pressure from } \mathcal{R} \setminus G, \\
E_{\text{flow}}(t) &= \text{instantaneous energy flux, and} \\
\alpha, \beta, \gamma, \delta &> 0 \text{ are system-specific coupling constants.}
\end{aligned}$$

Remark 4.2. The logarithmic term $\alpha \ln(1 + |E|)$ captures the diminishing marginal entropy contribution of each additional edge — analogous to the sub-additive behavior of Boltzmann entropy. The linear uncertainty and latent terms ensure that incomplete knowledge of the graph contributes positively to entropy, consistent with Shannon’s treatment of missing information.

4.3 Consistency with Classical Limits

When $U \rightarrow U_{\min}$ (near-complete knowledge), $L \rightarrow 0$ (no hidden causes), and energy flux reaches steady state, $S_{\mathcal{C}}$ reduces to a monotonically increasing function of $|E|$ alone. In a system with $|E|$ proportional to accessible microstates, this recovers the qualitative behavior of Boltzmann entropy.

5. Hidden Variable Theorem

Theorem 5.1 (Hidden Cause Entropy Lifting). *If a hidden causal node $H \notin G$ exists such that H influences observable dynamics, then:*

$$\Delta(\text{PredictionError}) > 0 \implies \Delta U > 0, \quad \Delta L > 0 \implies \Delta S_{\mathcal{C}} > 0.$$

Hidden causes manifest as entropy increase in the observable graph.

Proof. By Axiom 5, H influences observable nodes without appearing in G . An observer using G as their model will incur systematic prediction error. By definition of U (residual uncertainty after exhausting observable explanations), this error cannot be attributed to any $n \in N$, so U must absorb it: U increases. Simultaneously, the gap $|\mathcal{R} \setminus G|$ grows, increasing latent pressure L . By Definition 4.1, both terms contribute positively to $S_{\mathcal{C}}$. $\square$

Corollary 5.2. *Entropy growth in a closed observable system is not only the result of increasing internal complexity, but also of unresolved external causal influence.*

Systems that appear to be in isolated equilibrium may be experiencing slow hidden-variable entropy injection.

6. Entropy Growth Law

Proposition 6.1 (CEH Second Law Analog). *The rate of causal entropy growth is:*

$$\frac{dS_C}{dt} = \alpha \frac{d|E|}{dt} + \beta \frac{dU}{dt} + \gamma \frac{dL}{dt} + \delta \left| \frac{dE_{flow}}{dt} \right|. \quad (4)$$

Each term is independently non-negative under the following conditions:

- (i) $d|E|/dt \geq 0$: *interactions do not vanish once realized (causal irreversibility),*
- (ii) $dU/dt \geq 0$: *uncertainty does not spontaneously resolve without new observations,*
- (iii) $dL/dt \geq 0$: *hidden structure does not spontaneously become observable,*
- (iv) $|dE_{flow}/dt| \geq 0$: *by definition.*

Therefore $dS_C/dt \geq 0$, in agreement with the classical second law.

Remark 6.2. This constitutes a *causal derivation* of entropy non-decrease. The classical second law is recovered not from phase-space counting arguments but from the irreversibility of causal edge formation.

7. Equilibrium Redefined

Classical thermodynamics defines equilibrium as the state of maximum entropy, beyond which no further change occurs. CEH modifies this as follows.

Definition 7.1 (Causal Equilibrium). A system is in **causal equilibrium** when:

$$\frac{dS_C}{dt} \rightarrow 0, \quad \text{while} \quad U > 0.$$

Entropy growth becomes negligible, but uncertainty remains. Perfect equilibrium (i.e., $U = 0$) is excluded by Axiom 6.

This redefinition has a physical consequence: no system ever fully “dies.” Residual uncertainty guarantees non-zero latent pressure $L > 0$, which can seed future edge formation. The apparent endpoint of thermodynamic evolution is, under CEH, a metastable plateau rather than a true terminus.

7.1 CEH Equilibrium vs. Thermodynamic Maximum Entropy

A referee familiar with classical thermodynamics will immediately raise the following objection:

Standard thermodynamics identifies equilibrium with maximum entropy. If the initial universe was in equilibrium, its entropy should have been high, not low. CEH cannot simultaneously claim a compressed initial state and call it equilibrium.

This objection is valid against classical thermodynamics but does not apply to CEH, because CEH redefines equilibrium. The distinction is critical and must be stated precisely:

	Classical Thermodynamics	CEH
Equilibrium condition	$S = S_{\max}$	$dS_C/dt \rightarrow 0$
Entropy at equilibrium	Maximum	Any value
Uncertainty at equilibrium	Zero (in principle)	$U > 0$ always
Is equilibrium a terminus?	Yes	No

In CEH, equilibrium is a *rate condition*, not a *value condition*. A system is in causal equilibrium when its entropy is no longer growing, not when its entropy has reached a maximum. A newly formed, highly compressed state — with minimal realized interactions — satisfies $dS_C/dt \approx 0$ trivially: there is nothing yet to interact, so nothing is growing. This is a causal equilibrium of minimal entropy, not maximal entropy.

The initial universe under CEH is therefore interpreted as a system in *causal equilibrium at minimum entropy*: no differentiated cause–effect structure had yet emerged, no energy redistribution had occurred, and the entropy growth rate was effectively zero not because the system was exhausted, but because causal unfolding had not yet begun.

8. Emergent Compression and the Super-Node Principle

8.1 Emergent Compression Principle

Proposition 8.1 (Graph Compression). *When a causal graph G reaches near-equilibrium — formally, when $dS_C/dt < \varepsilon$ for a sustained duration τ — the graph undergoes compression:*

$$\mathcal{C}(G) = N_{\text{super}},$$

where N_{super} is a single higher-level node whose internal structure encapsulates the full causal history of G .

Intuitively: a causal graph that has stabilized — whose components have organized into a self-consistent structure — becomes, from the perspective of the next level of description, a *single entity*.

8.2 Super-Node Expansion

Proposition 8.2 (Causal Reseeding). *A super-node N_{super} , once formed, acts as an atomic unit in a new causal graph G_{new} :*

$$N_{\text{super}} \longrightarrow G_{\text{new}}.$$

New interactions involving N_{super} begin to form edges in G_{new} , initiating a new epoch of causal unfolding.

8.3 Recursive Reality Principle

Combining Propositions 8.1 and 8.2:

$$G_0 \longrightarrow N_1 \longrightarrow G_1 \longrightarrow N_2 \longrightarrow G_2 \longrightarrow \cdots \quad (5)$$

Reality self-organizes recursively. Each stable causal graph becomes an entity at the next level of description. This pattern is observed empirically across scales: particles organize into atoms, atoms into molecules, molecules into cells, cells into organisms, organisms into societies. CEH provides a single causal mechanism for this ubiquitous phenomenon.

8.4 Level-Relative Entropy

The recursive structure (5) has an immediate consequence for the interpretation of entropy across levels.

Proposition 8.3 (Non-Transfer of Maximum Entropy). *Maximum causal entropy at level n does not imply maximum causal entropy at level $n + 1$:*

$$S_{\mathcal{C},n} = S_{\mathcal{C},n}^{\max} \not\Rightarrow S_{\mathcal{C},n+1} = S_{\mathcal{C},n+1}^{\max}.$$

When G_n compresses to N_{n+1} , the entropy of the new graph G_{n+1} resets to a low value: the super-node N_{n+1} has at most zero realized edges in G_{n+1} at the moment

of its formation.

Sketch. By Proposition 8.1, G_n at near-equilibrium compresses to N_{n+1} . At the moment of compression, N_{n+1} is a single node in G_{n+1} with $|E_{n+1}| = 0$. By Definition 4.1 and Hypothesis 3.3, $S_C^{(n+1)} \approx S_{\min}$ at this moment, regardless of the value of $S_{C,n}$. $\square$

This proposition has a direct cosmological interpretation. If the initial universe is interpreted as a super-node formed from the compression of a prior causal level, then its minimal entropy is not puzzling: it is the *expected entropy of a freshly formed super-node in a new causal graph*.

An observer embedded within G_n would perceive that graph as having reached maximum entropy (local equilibrium). But from the perspective of G_{n+1} , the same configuration is a single compressed node at minimum entropy — ready to begin a new epoch of causal unfolding.

8.5 Formal Requirements for Inter-Level Entropy Transfer

For Proposition 8.3 to constitute a full theory rather than a structural sketch, the following must be formally defined:

- (i) **The compression operator $\mathcal{C}$:** a precise mathematical map from a near-equilibrium graph G_n to a super-node N_{n+1} , specifying what causal information is preserved and what is coarse-grained.
- (ii) **The entropy reset rule:** a formula specifying $S_C^{(n+1)}$ at the moment of super-node formation as a function of $S_{C,n}$ and the compression operator.
- (iii) **The seeding mechanism:** a specification of how N_{n+1} begins to acquire edges in G_{n+1} — i.e., what triggers the first realized interaction at the new level.
- (iv) **The level metric:** a definition of what constitutes “level n ” versus “level $n + 1$ ” in physical terms, so that the hierarchy is empirically grounded rather than formally arbitrary.

These are open problems. Their resolution would transform the Recursive Reality Principle from a qualitative framework into a quantitative theory.

9. The First Interaction Problem

Even granting that $I(0) \approx 0$ and $|E| \approx 0$ at the initial epoch, a fundamental question remains:

If the universe began in causal equilibrium with no realized interactions, what produced the first edge? Why did causal unfolding begin at all?

9.1 Why Uncertainty Cannot Answer This

A natural candidate answer uses Axiom 6: since $U > 0$ always, perfect stasis is impossible, so a fluctuation must eventually occur, producing the first interaction. This argument appears in some earlier formulations of CEH but is *invalid within the framework*, for a reason made explicit in the scope condition of Axiom 6.

Uncertainty U is an observer-dependent, operationally defined quantity. It requires: (i) an observer, (ii) observations, (iii) data. By Axiom 2, data requires prior observations, observations require prior effects, and effects require prior interactions. Therefore at $I = 0$ — before any interaction has been realized — there is no observer, no observations, no data, and U is not operationally defined.

Invoking $U > 0$ to produce the first interaction is circular: it uses a quantity whose definition presupposes interactions to explain how interactions begin.

9.2 The Recursive Reality Resolution

CEH offers a different resolution that does not require an unexplained first cause: *there may be no absolute first interaction.*

Under the Recursive Reality Principle (5), reality organizes into an unbounded recursive hierarchy of causal levels:

$$\cdots \longrightarrow G_{n-1} \longrightarrow N_n \longrightarrow G_n \longrightarrow N_{n+1} \longrightarrow G_{n+1} \longrightarrow \cdots$$

What appears to be the “initial state” of any observable level G_n is the super-node N_n — the compressed output of the prior level G_{n-1} reaching equilibrium. The “first interaction” at level n is therefore caused by dynamics already present at level $n - 1$: specifically, the formation of N_n and its contact with other nodes at the new level.

The full chain at level n is:

$$G_{n-1} \xrightarrow{\text{equilibrate}} N_n \xrightarrow{\text{enter } G_n} N_n + N'_n \xrightarrow{\text{first edge}} |E_n| > 0 \longrightarrow \frac{dS_{\mathcal{C},n}}{dt} > 0 \longrightarrow \text{causal graph growth.} \quad (6)$$

The regress does not terminate at an uncaused first cause. It continues upward through the recursive hierarchy. At each level the “first interaction” has a causal explanation from the level above. No level requires a causeless origin.

9.3 Scope of the Resolution

This resolution has structural appeal — it dissolves the first-cause problem rather than punting it to a boundary condition. Several honest caveats apply:

- (i) The recursive hierarchy is a structural feature of CEH, not derived from the axioms alone. A formal proof that every equilibrated graph *must* produce a super-node entering a new graph has not been given.
- (ii) The resolution replaces a singular uncaused beginning with an infinite recursive structure. Both positions are presently unfalsifiable at cosmological scales; the recursive resolution is preferred on internal consistency grounds within CEH, not on empirical grounds.
- (iii) Crucially, the cause of the first edge at level n — the dynamics of G_{n-1} — is unobservable from within G_n . It is, precisely, a hidden variable in the sense of Axiom 5. By Theorem 5.1, this manifests as an entropy increase from an apparently uncaused source — which is exactly the phenomenology of a Big Bang as observed from within the current causal level.

10. Application: Black Hole Entropy

10.1 Classical Result

The Bekenstein–Hawking entropy of a black hole of horizon area A is [3, 4]:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A.$$

Its proportionality to *area* rather than volume is one of the most surprising results in theoretical physics, motivating the holographic principle [5, 6].

10.2 Causal Interpretation

Under CEH, a black hole is an *extreme causal convergence region*: a domain in which all causal trajectories that enter converge and cannot exit. Define the path convergence at a point x as:

$$\mathcal{P}(x) = \sum_i \text{Path}_i(x),$$

where the sum runs over all causal paths passing through x .

Hypothesis 10.1 (Black Hole Causal Entropy). The entropy of a black hole is proportional to the total causal path convergence at its boundary:

$$S_c^{\text{BH}} \propto \oint_{\partial\mathcal{H}} \mathcal{P}(x) dA,$$

where $\partial\mathcal{H}$ is the event horizon.

This recovers the area scaling: the horizon is precisely the surface at which causal paths converge maximally, and its area counts the number of such convergence events. The holographic bound arises naturally as the limit on causal path density at the horizon surface.

11. Testable Predictions

CEH makes the following falsifiable predictions:

- P1. Hidden-Cause Entropy Injection.** In any modeled system, the deliberate injection of unobserved hidden causes (latent variables) should produce a measurable increase in prediction error, residual uncertainty U , and therefore in any entropy proxy of the system — independently of whether the system’s energy flux changes.
- P2. Graph Saturation Entropy Plateau.** In network systems where the causal graph approaches completion (i.e., $d|E|/dt \rightarrow 0$), entropy growth should plateau even if energy redistribution continues. This is testable in discrete simulations of interacting-agent networks.
- P3. Super-Node Compression.** In sufficiently stable causal graphs — biological neural networks, social institutions, crystalline structures — compression into effective higher-level units should be observable as a reduction in the description-length required for the system, followed by renewed complexity growth as the

compressed unit enters new interactions.

P4. Scale Invariance of Causal Pattern. The recursive structure (5) predicts self-similar causal organization across scales. Specifically, the entropy growth rate dS_C/dt , when measured using appropriate scale-invariant proxies, should exhibit the same functional form at particle, organism, social, and cosmological scales.

12. Discussion

12.1 Relation to Classical Thermodynamics

CEH does not contradict classical thermodynamics. The second law is recovered in Proposition 6.1 from a deeper causal argument. The Boltzmann and Shannon entropies remain valid as special cases.

Regarding the initial low-entropy condition, the framework’s explanatory reach must be stated precisely. CEH contains an *internal* answer to the question “why was the causal graph empty at $t \rightarrow 0$?”: it was empty because no energy redistribution had yet been realized ($I \approx 0$), and by Axiom 2 no causal edges can form without realized interactions. The empty graph is derived, not assumed (see derivation chain (2)).

What CEH does *not* yet derive is the more primitive question: why was the initial state so highly constrained? This remains an open assumption. A critic may press the regress one level further — “why had no energy redistribution yet occurred?” — and at that level CEH currently has no answer beyond the postulate of an initial constrained state. This is not a deficiency unique to CEH. Standard cosmology also accepts the initial low-entropy condition without deriving it; Penrose [1] treats it as a boundary condition on the wavefunction of the universe, and Carroll and Chen [2] treat it as a statistical fluctuation in a larger ensemble. CEH adds a third option: the constrained initial state as the natural zero-point of a causal graph whose unfolding *is* thermodynamic history. Whether this zero-point can be grounded in a still-deeper principle is a direction for future work.

12.2 Proof of Concept: Causal Discovery Without Ground Truth

The core epistemic claim of CEH — that causal structure, latent variables, and uncertainty can be tracked and measured by a finite observer with no access to the underlying ground-truth equations — has a working empirical demonstration.

The Engine Behavior Intelligence System (EBIS) is a causal intelligence platform built for internal combustion engine analysis. The system operates on live engine telemetry without knowledge of the engine’s governing equations or physical parameters. From raw cycle data alone, EBIS incrementally constructs a causal graph G_t , flags edges exhibiting latent variable signatures (unmeasured confounders), maintains an explicit uncertainty measure (per-edge trust), and grows a knowledge-value metric as causal edges accumulate across experimental runs.

This constitutes a concrete existence proof for Axioms 5 and 6: a finite observer, embedded in a system it cannot fully observe, can recover meaningful causal structure while maintaining honest quantification of what remains unknown. The system does not converge to certainty; it converges to a stable uncertainty floor — precisely the behavior predicted by Definition 7.1. Formalizing the correspondence between EBIS’s runtime quantities and S_C is a direction for future work.

12.3 Relation to Causal Set Theory

The ontological commitment of CEH — that causal graphs are more fundamental than states — resonates with causal set theory (CST) [7, 8], in which spacetime is replaced by a discrete partially ordered set of events. CEH extends this intuition from spacetime structure to entropy itself, proposing that the causal record is not merely a description of spacetime but the substrate of thermodynamic evolution. A formal connection between CEH and CST is an important direction for future work.

12.4 Relation to Information-Theoretic Entropy

Jaynes’ maximum entropy principle [9] treats entropy as an epistemic quantity: it measures what an agent does not know. CEH agrees that uncertainty U contributes to entropy (the βU term in Definition 4.1), but insists that entropy also has an *ontic* component: the realized causal edges $|E|$ and energy flux term represent actual physical history, not merely epistemic deficit.

12.5 Limitations

The present framework is conceptual and semi-formal. Several open problems remain:

- Precise operational definitions of U and L in physical systems.
- Derivation of the coupling constants $\alpha, \beta, \gamma, \delta$ from first principles or from empirical fitting.

- A rigorous proof that S_C reduces to S_{BH} in the black hole limit.
- Quantitative reconciliation with quantum statistical mechanics.

13. Conclusion

We have presented the Causal Entropy Hypothesis: a graph-theoretic framework in which entropy is the realized causal unfolding of energy. Beginning from five explicit axioms, we derived:

- A composite causal entropy functional S_C (Definition 4.1).
- A causal derivation of entropy non-decrease (Proposition 6.1).
- A Hidden Cause Entropy Lifting theorem (Theorem 5.1).
- A causal equilibrium condition compatible with irreducible uncertainty (Definition 7.1).
- An Emergent Compression Principle and recursive hierarchy structure (Propositions 8.1, 8.2).
- A causal interpretation of black hole entropy.

The central resolution of the low-entropy initial condition puzzle is: *the universe began with low entropy because it began with almost no realized causal interactions — the causal graph was nearly empty.* This is not a special initial condition requiring explanation; it is the only consistent starting point for a universe whose causal structure is itself the substance of thermodynamic evolution.

Whether this framework ultimately deepens our understanding of thermodynamics or leads to empirically distinguishable predictions at physical scales remains to be seen. We offer it as a conceptual foundation and invite formal engagement.

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[To be completed by author.]

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